

CH5440 Multivariate Data Analysis

*We are drowning in information
and
starved for knowledge*

Tom Peters in his book on “Thriving on
Chaos”

Harper Perennial, 1991
Edition : Reprint
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ISBN-13 : 978-0060971847

Resources

Textbooks:

1. Maiti J, Multivariate Statistical Modeling in Engineering and Management, CRC Press, 2023
2. **Backhaus K, B Erichon, S. Gensler, R. Weiber, T. Weiber, Multivariate Analysis – An Application Oriented Introduction, 2nd ed., Springer Gabler, 2023**
3. Johnson R A and D W Wichern, Applied Multivariate Statistical Analysis, Prentice Hall, 2002
4. Montgomery, D C and G C Runger, Applied Statistics and Probability for engineers, Wiley, 2003
5. Joseph F H, W C Black, BJ Babin and R E Anderson, Multivariate Data Analysis, 8th ed., Cengage, 2019
6. Afifi, A., May, S., Donatello, R., & Clark, V.A. (2019). Practical Multivariate Analysis (6th ed.). Chapman and Hall/CRC. <https://doi.org/10.1201/9781315203737>
7. <https://stats.oarc.ucla.edu/other/dae/>
8. <https://www.routledge.com/Practical-Multivariate-Analysis/Afifi-May-Donatello-Clark/p/book/9781032088471>
9. Any reference inadvertently omitted will be added on intimation

Multivariate Data Analysis (MVDA): Introduction

- ❖ Analyses of data containing many features (predictors, independent variables) and numerous observations (responses, targets) often running into several thousands

For e.g.,

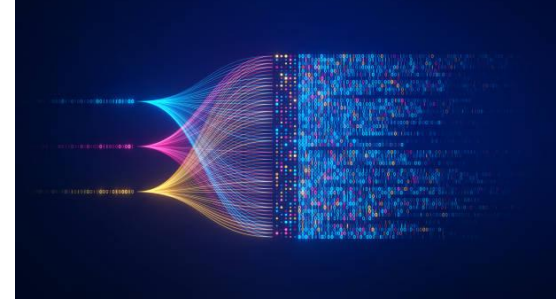
- A medical data base to evaluate liver parameters of young patients
- To evaluate the financial state of a company, the intelligent investor examines 5-10 indexes of the company's performance in that financial year

Multivariate Data Analysis (MVDA): Introduction

- ❖ Usually, the features/data to at least some of these queries are interrelated/correlated/missing
- ❖ Recognizing complicated interrelationships between various features & interpreting these results makes MVDA a highly profitable exercise

Often results and insights are obtained that may not have been attained without multivariate analysis

Big Data & Its Management



- ❖ Explosion in data today
- ❖ **Varied Sources:** Social media and online behavioral patterns;
- ❖ **IoT** has brought connectivity to almost every type of device;
- ❖ Substantial amount of data in genomics, neuroscience and astrophysics;
- ❖ Storage devices have capacity to hold Big Data (cloud, servers, data warehouses) and software to manage them
- ❖ Companies study and understand their customers through data collected from them/associated with them,
- ❖ This data help **informed decision-making** leading to **higher profits**

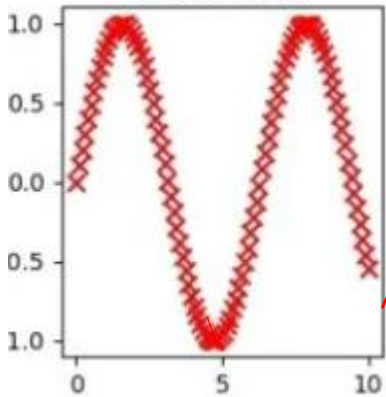
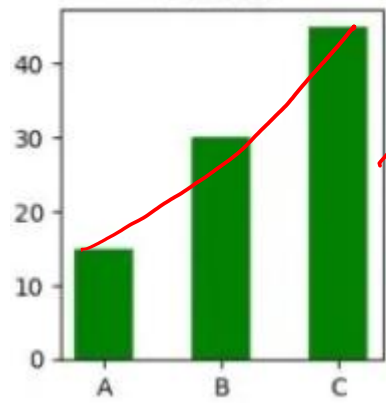
Points to Heed When Collecting Data

- ❖ **Define goals:** Purpose of collecting data
- ❖ **Validate :** Data clean, without outliers and missing elements?
- ❖ **Analyze:** From the data identify trends, patterns, and correlations
- ❖ **Visualize:** Plot the data - box plots, scatter plot matrices, three-dimensional contours
- ❖ **Inferring the Data:** How may the information content in the data collected be streamlined to optimized marketing strategies?

Handwritten notes and diagrams:

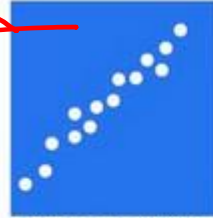
- Age ↓
20
25
30
10²
- Income
500000
600000
10⁶
- Normalizing
Age - Age
5 Age
- data cleaning
labels 1 2 3
Nominels
- .5 -1
-1.2 -1.5
0 0.5
0.7 0





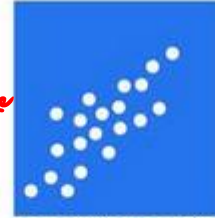
Trend - I
Pattern - II
Correlation - III

Comfort level



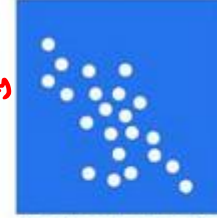
Strong positive correlation

hours of study



moderate positive correlation

Degree



Moderate negative correlation

age



Strong negative correlation

Temperature



No correlation

Gender



Curvilinear relationship

Covariance
Correlation

Statistical analysis of data

Mean

Cov

Corr

Statistical dist.

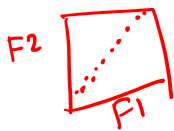
Multivariate

Normal

distn.

$$\begin{bmatrix} 1 & 1.1 & \dots \\ 2 & 1.9 & \dots \\ 3 & 2.99 & \dots \end{bmatrix}^{-1}$$

NaN

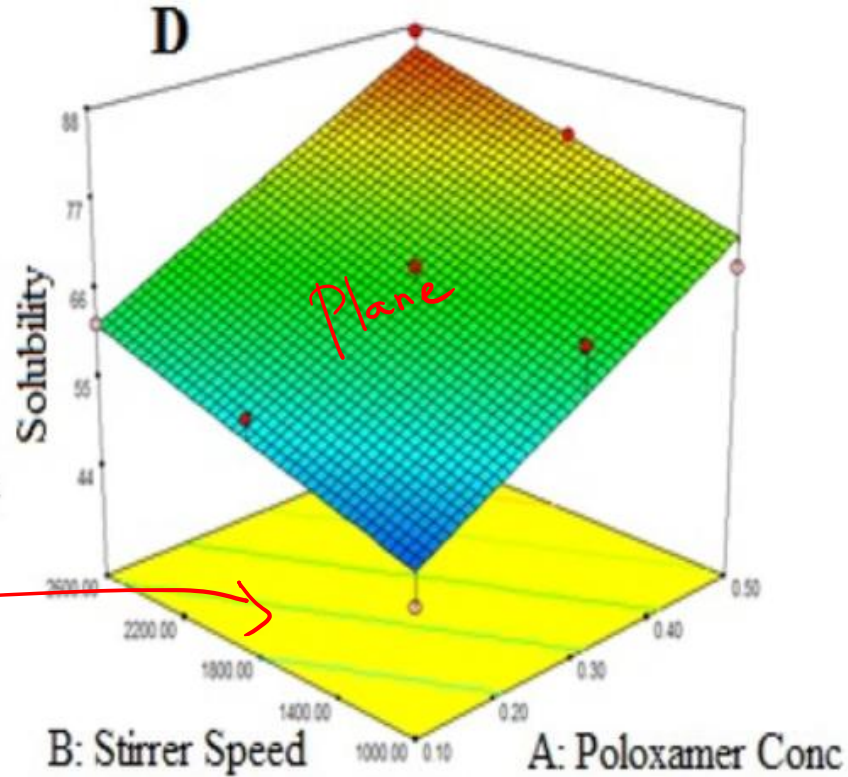
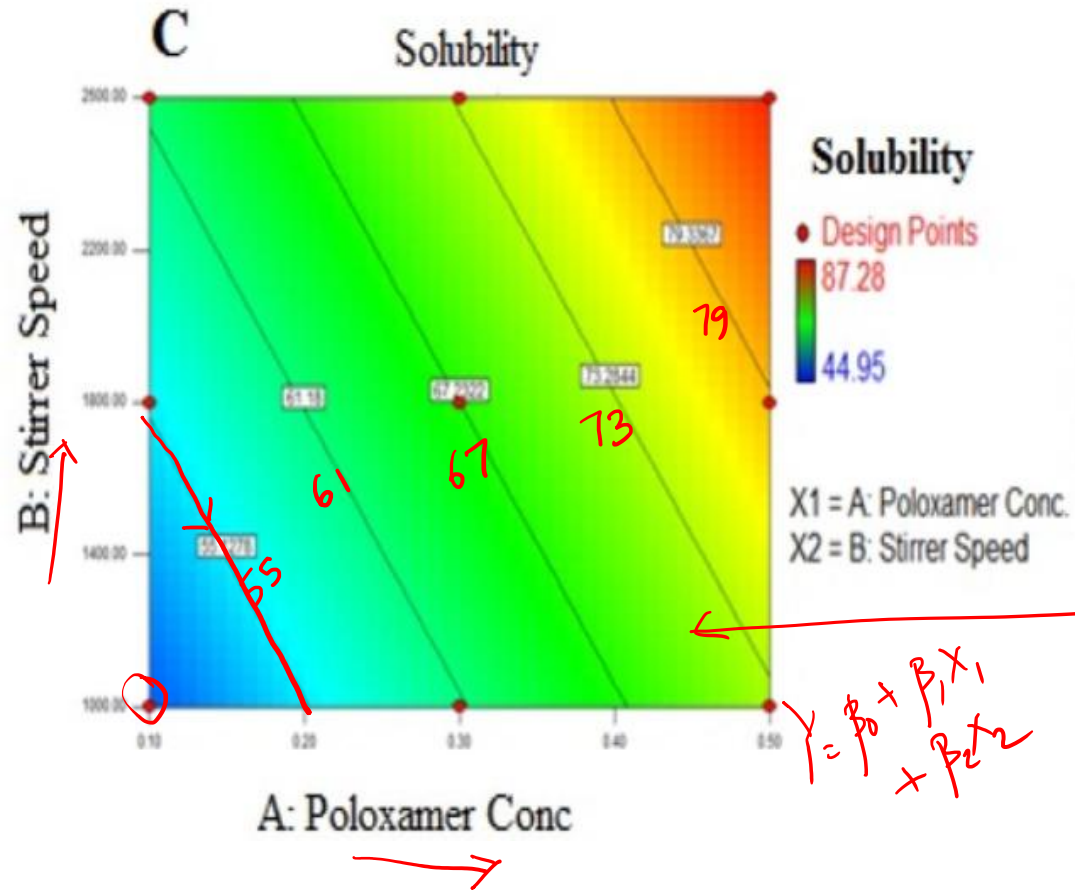


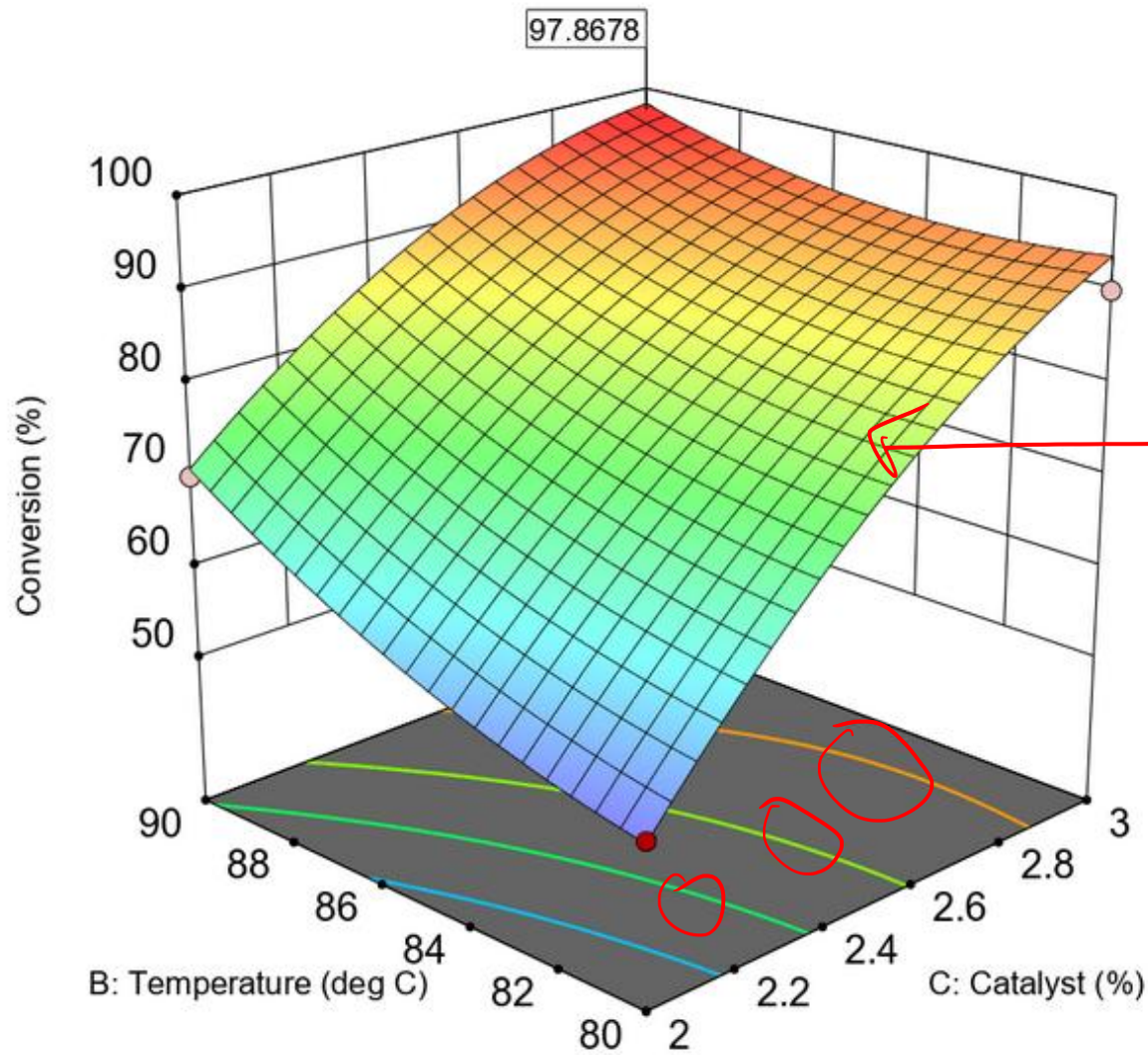
F1 F2 ... F100
F2 & F3

III

<https://medium.com/@boukamchahamdi/key-data-visualization-techniques-for-understanding-trends-and-patterns-1bce981d408f>

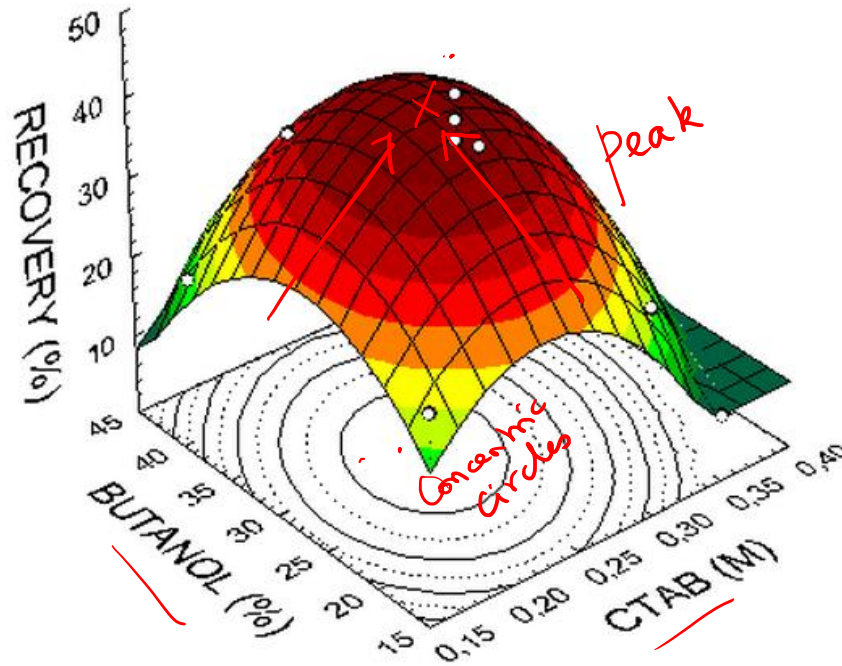
<https://www.facebook.com/AnalyticsVidhya/posts/pearson-correlation-coefficient-r-often-denoted-as-r-measures-the-strength-and-d/967997301998287/>





$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$

Curvature



Handwritten notes and equations:

A 2D graph showing a parabolic curve with a peak labeled y_{max} at x .

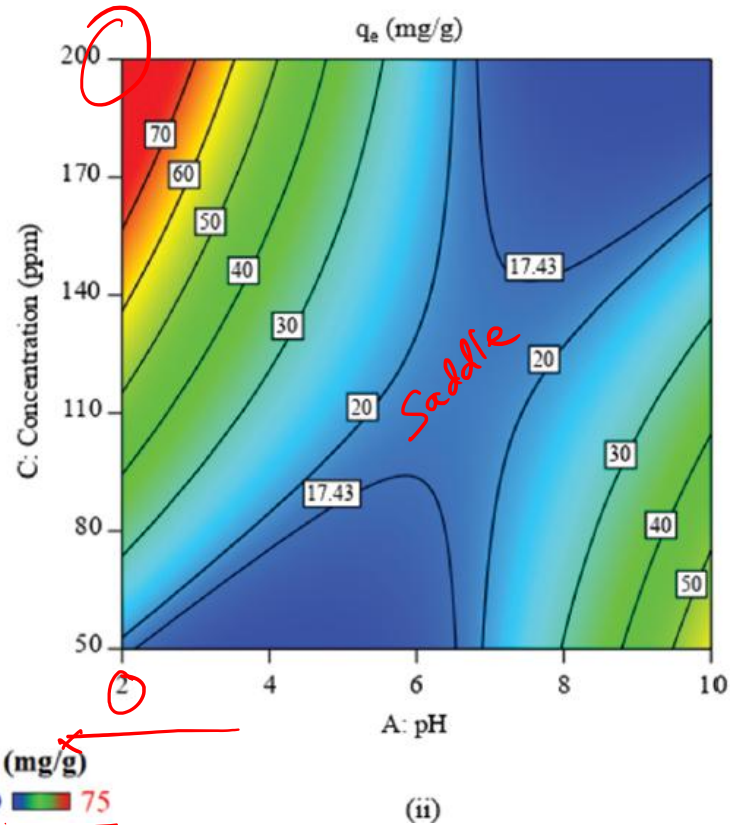
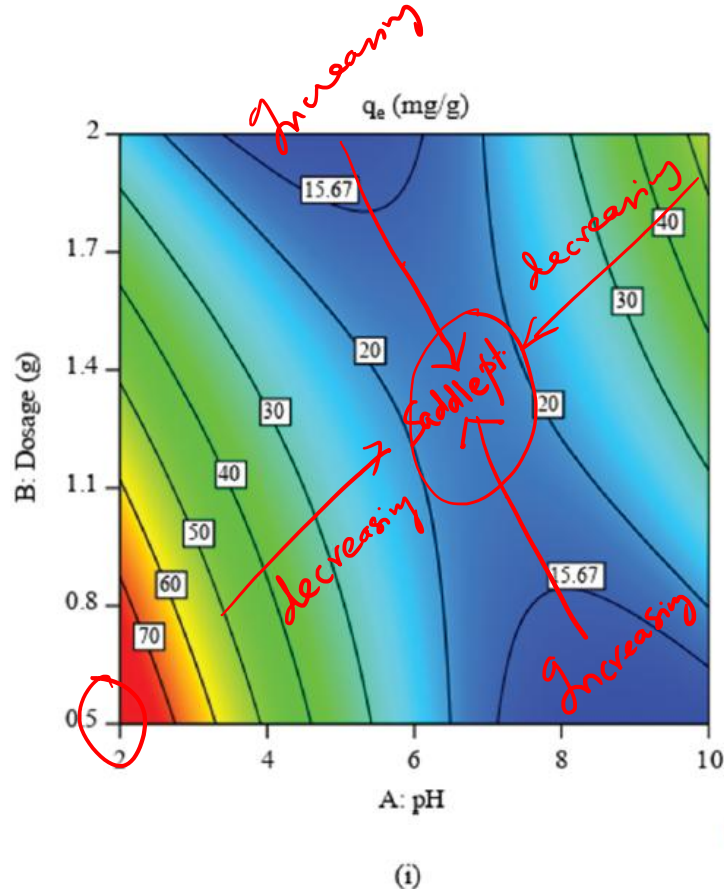
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2$$

Quadratic peaks & valleys

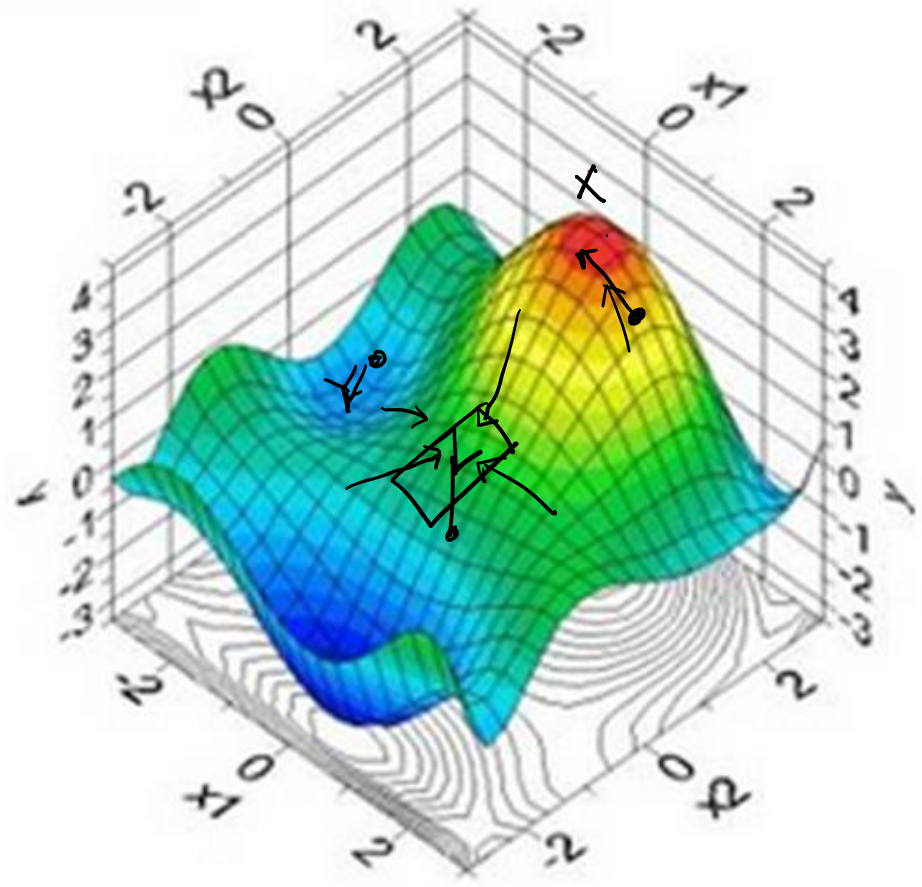
Figure 1. Response surface described by the model, which represents β -xylosidase recovery as a function of CTAB and butanol concentration.

Extraction conditions: pH 8.0, hexanol concentration 10%, electrical conductivity 4 mS/cm, temperature 26°C, CTAB concentration 0.26M and butanol concentration 29.4%.

Presentation of contours, surfaces and 3-D plots



c



f_{min} Search

Obj. Min.

- Obj. Max.

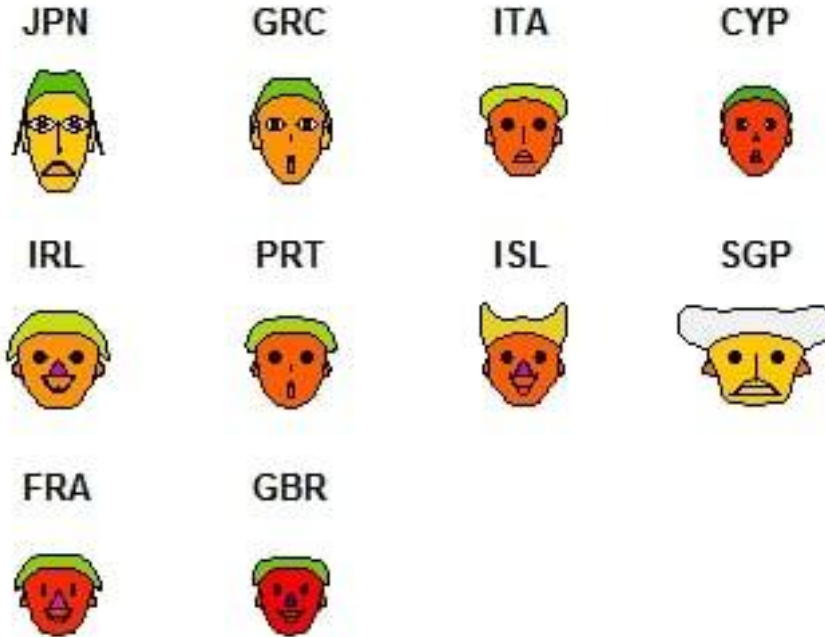
Starting Guesses

Trends and Patterns

Analyze: From the data identify trends, patterns, and interrelationships

- ❖ A trend is the overall direction or long-term movement of a variable over time (e.g., an upward or downward slope).
- ❖ Pattern is a specific, repeating design, sequence, or recurring behavior within that data (e.g., daily dips/seasonal spikes or sinusoidal variation)
- ❖ A trend dictates the overall direction you are heading, while
- ❖ A pattern is the predictable rhythm of events along that journey

A comparative view using Chernoff faces



Source : www.worldbank.org

Big Data in (Chemical) Engg.

- ❖ Improve productivity, reduce toxic effluents (zero liquid discharge and recycle) and be sustainable
- ❖ Facilitate real-time decisions on production scheduling & use of raw materials
- ❖ **Preventative maintenance:** Sensors fixed on equipment collect data. Patterns are identified and predictions may be made as to when the equipment might begin to show symptoms of malfunctioning;
Advantage: Reduced downtime
- ❖ **Quality control:** Big data analytics help engineers ensure product quality retention and improvement

Big Data in (Chemical) Engg.

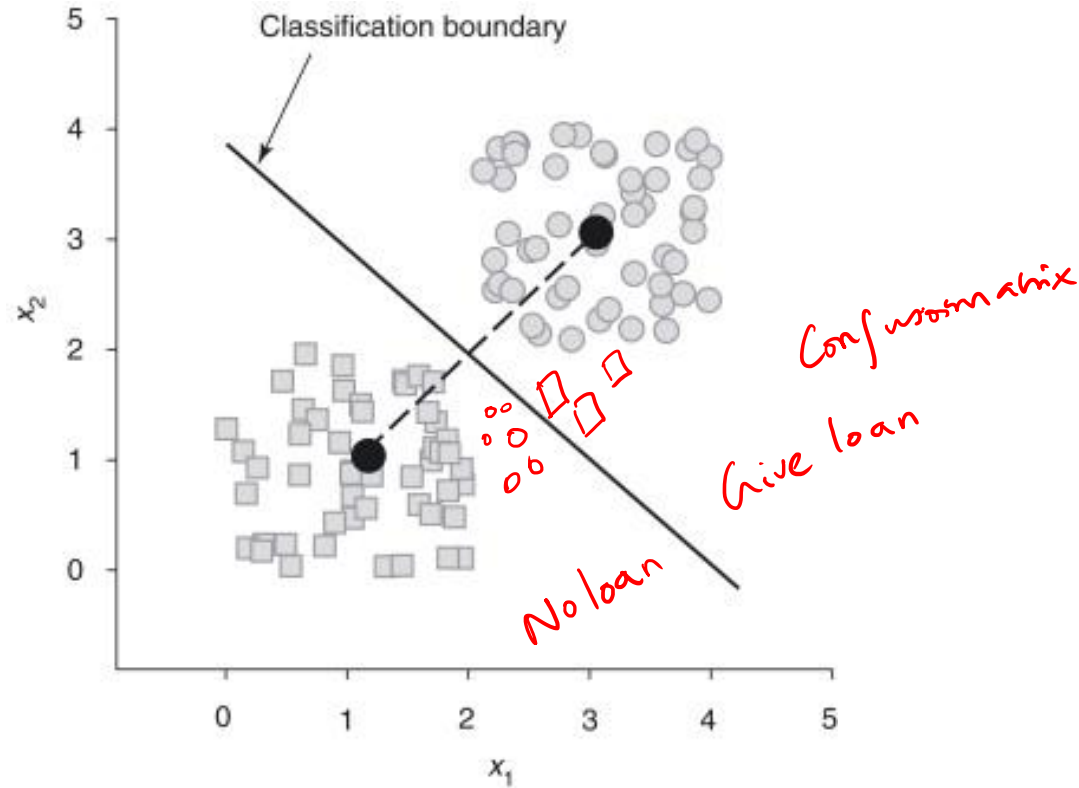
- ❖ **Supply chain management:** Optimize inventory levels
- ❖ **Machine learning:** Automate tasks
- ❖ **New product development:**
 - New drugs,
 - Alternate pathways for products and processes (new ways of manufacturing), and
 - clean energy solutions

Topics in MVA: Multiple Linear Regression

- ❖ An engineer determines variables to maximize the extraction of a medicinal compound from an herb (Speciality Chemical)
- ❖ Data on effect of **plant leaf size**, hot water temperature, **hot water amount**, **time and level of stirring**, have been collected.
- ❖ A multiple linear regression equation is computed, where % Medicine Extracted is the outcome variable, while the other factors are predictors
- ❖ A linear combination of the predictors is used to develop a multivariate linear regression model to predict and optimize the extraction of the herb



Topics : Discriminant Function Analysis (DFA)



<https://www.sciencedirect.com/topics/neuroscience/discriminant-function-analysis>

Topics : Discriminant Function Analysis (DFA)

- ❖ Used when the independent variables are metric
- ❖ A method for analyzing differences between two or ^{more} groups,
- ❖ For e.g., how the voting behavior for political parties differs in terms of voters' socio-demographic and psychographic characteristics (attitudes, lifestyles, values, personalities, interests) i/o age and income
- ❖ The dependent variable identifies the group membership, that is, the chosen political party, and
- ❖ The independent variables describe the group elements, that is, the voters' socio-demographic and psychographic characteristics.

Discriminant Function Analysis (DFA)

- ❖ A large sample of initially disease-free men greater than 50 yrs of age from a locality was tracked to see **who subsequently suffered from heart attack**.
- ❖ During the first survey, blood sample was taken from each man, & numerous other tests were conducted - BMI, cholesterol levels, lipids profile, and blood sugar
- ❖ Investigator determines a linear function of these to **predict** who would/who would not suffer from a heart attack within the next 10 years. Obviously, these would be a new set of patients.

DFA in Classification

- ❖ Once relationships between group membership and characteristics have been analyzed for a given set of subjects
- ❖ We may predict the group membership for 'new' subjects

Frequently implemented in

- ❖ Credit scoring (i.e., risk classification of bank customers applying for a loan)
- ❖ Safe or Unsafe Equipment depending on operating and geometric parameters

Backhaus K, B Erichon, S. Gensler, R. Weiber, T. Weiber, Multivariate Analysis – An Application Oriented Introduction, 2nd ed., Springer Gabler, 2023

Logistic Regression

Item purchased (1)
or not purchased (0)
 $= f$ [Income, Gender]
num. 0 Male
1 Female
M F

Linear Regression
%. Medicine
extracted
T, S, t

- ❖ Netflix/Prime etc., classifies movies into two different groups according to attraction to a high proportion or only to a low proportion of the viewing audience when the movie is streamed
- ❖ The company records data on features such as the **movie duration**, the genre (**action, romance, comedy**), and the **actors**.
- ❖ So, variables need not be only numerical but may be categorical as well
- ❖ From **logistic regression** we derive an equation to **estimate the probability** of capturing a **high proportion of the target audience**

y values i/p's 0 1 0
logistic Reg. y values o/p's 0.1 0.7 0.8
0.25

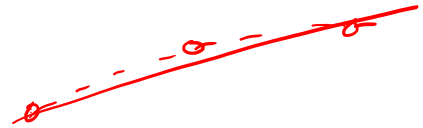
0.4 $p^* > 0.5$ $\boxed{\frac{1}{0} \checkmark}$
 $p^* < 0.5$



Caution on Models

Model Assumptions

- ❖ If the modeling results are satisfactory (e.g., good model fit, realistic model parameters) the researcher move to the next step
- ❖ Make inferences to understand the process being examined
- ❖ But researcher must recognize that any conclusions are tied to the proposed model
- ❖ Model may not be complete representation/description of the underlying process.



Caution on Models

- ❖ If the model incorrectly represents the process, inferences may be flawed (Science is replete with theories that were later disproved)
- ❖ Researcher must be aware that there may be many “alternative” models that may also represent the process but provide different conclusions.
- ❖ A model is sensitive to the range of features, interaction between them, nonlinearity of the actual model or model parameters drifting, and changing with time

$$\text{Medicine} = \text{Intercept} + \text{Shr. } \beta_1 + \beta_2 T + \beta_3 t$$

$-\beta_{12} T \times t$

Types of Variables: Manifest Vs. Latent

- ❖ **Manifest** (clear, evident, in plain sight) variables may be directly observed or measured (e.g., weight, height, hair color, age, price, quantities, and income).
- ❖ However, **latent** variables cannot be directly observed
- ❖ Examples of hypothetical variables are friendship, rivalry, trust, motivation, intelligence, and potential

Quantifying Latent Variables

- ❖ To measure latent variables, suitable operations involving manifest variables are necessary
- ❖ **Compositional Approach:** This implies that the value of a latent variable is composed of observations involving various manifest variables
- ❖ Intelligence, is often measured by a combination of various observed variables through standardized tests assessing cognitive abilities (reasoning, comprehension, and problem-solving), yielding an IQ score.

Types of Data: Cross-sectional

- ❖ Data are the 'raw material' to be processed by multivariate data analysis
- ❖ Cross-sectional data are collected by observing many different subjects or entities at a single point (snapshot) in time
(Annual Profits, Type of Service, Years of Operation, Number of Employees of different companies on 31.3.26)

Examples of Cross-sectional Data

- ❖ **Comparison** of the financial statements of companies at a fixed date
- ❖ **Diabetic levels:** A sample of 1,000 people randomly drawn from a population, with their weight, diet habits and exercise hours measured to calculate the percentage of the sample that is diabetic
- ❖ **Price/earnings ratios:** The price-to-earnings ratios of all BSE stocks on a particular date

Cross Sectional Data Spreadsheet

Firm	Annual Salary (Lakhs)	Service Type	Major Gender Hired	HQ Location
A	50	Public Sector	M	North
B	60	Health Care	F	West
C	40	Academics	M	South
D	90	Private Sector	F	East

Hiring of Personnel in 2025

Types of Data: Time Series

- ❖ Time Series data (longitudinal data) are collected at regular time intervals (e.g., CO₂ emissions, monthly sales, raw materials intake).
- ❖ Time Series Data measure how a variable (d)evolves over time.
- ❖ Time series data have a natural temporal ordering

Examples of Time Series Data

- ❖ Temperatures and seasonal rainfall measured monthly over a period of many years, will be helpful to predict future weather and climate change and consequently agricultural output
- ❖ **Financial data:** Stock prices, interest rates, and industry forecasts published at timely intervals are all examples of financial time series data.

Examples of Time Series Data

❖ **Sensor data:** Monitoring of heavy machinery to forecast maintenance issues in manufacturing and industrial settings

Real-time data used to identify potential failures before they cause sudden, expensive failures.

Continuously tracking vibration, temperature, pressure etc., enable transition from **reactive repairs** to **proactive maintenance**.

PREDICTIVE MAINTENANCE

FIX BEFORE FAILURE



4:26

Examples of Both Cross-sectional and Time Series Data

Data may be time series as well as cross-sectional when it shows both the characteristics of a group at a specific point in time and also shows the variation in characteristics of that group over a period in time.

Examples:

- ❖ Opening prices data of 20 stocks every day for a year is both time series and cross-sectional data.
- ❖ Studying the GDP of three developing countries over a period of three years is a combination of time series and cross-sectional data
- ❖ 60 Students marks over the multiple evaluations in a semester

Types of Variables: Nominal

- ❖ Here, each observation belongs to one of several distinct categories.
- ❖ The categories may or may not be necessarily numerical
- ❖ For example, “Gender” is a nominal variable.
- ❖ An individual’s gender may be male (M) or female (F) category
- ❖ Number 7 on MS Dhoni’s vest is a nominal variable (without quantitative value, inherent order, or numerical hierarchy)

Player #	
10	7
Job	
Man	
Acad.	
Military	
Gender	
Label	Label
10	0
7	1
Gender	
M	0
F	1
Job type	
1	0
3	0

Types of Variables: Nominal

- ❖ In data analysis, numbers are used as symbols since many computer programs are designed to handle only numerical symbols.

0 blank
1

- ❖ For example, we may use 0 and 1 to represent males and females, respectively.

- ❖ We may also use 1 and 2 to avoid confusing zeros with blanks.

- ❖ Any two other numbers can be used provided they are used consistently.

Indicator / dummy
0 0
0 1
1 0

one hot encoding
1 0 0
0 1 0
0 0 1

Types of Variables: Nominal

Learn

- ❖ Different numbers should be assigned to different attributes and identical numbers to identical attributes (one-to-one correspondence).
- ❖ For nominal variables, **arithmetic operations** (such as addition, subtraction, multiplication, or division) are **NOT** permitted
- ❖ A mean value of 2.0 for the three numbers 1, 2 and 3 would not be meaningful.
- ❖ However, we can infer frequencies or percentages by counting how often the different attributes occur in a data set (e.g., 25% red; 40% yellow; 35% green).

*1
1
0
0* *#1 60%
p = 0.60*

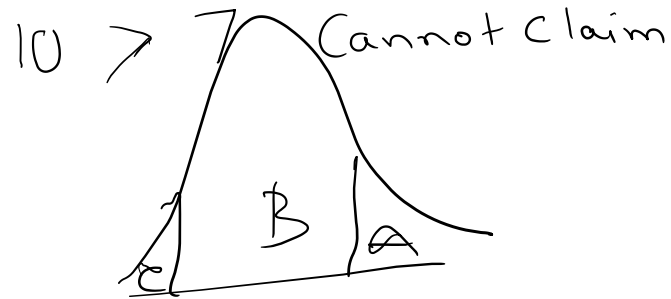
Eg of Nominal Variables

Rural

Urban

Blood groups - O +ve, A, B, B +ve

Location - Chennai, Mumbai, Kerala



Grades

$A > B$ (Nominal variable)

Fair < V. good < Excellent

Ordinal
variables

$$\text{Avg. grade} = \frac{A + B + C}{3}$$

Types of Variables: Ordinal

- ❖ Categories are used for ordinal variables as well, but there also exists a known order/hierarchy (junior-senior or Fair/Good/Excellent) among them.

fair Good V. good

- ❖ Ordinal data contain more information than nominal data, as they reveal order

Labels 1 2 3
1 < 2

Jan Fes March

Nominal
Variable

- ❖ Ordinal data are also referred to as **rank order data**.

Dy. Manager Manager Sr. Manager Cold Warm Hot Most polluted clean

Types of Variables: Ordinal

❖ For e.g., in the Mohs Hardness Scale, minerals and rocks are classified according to ten levels of hardness.

❖ The hardest mineral is diamond while the softest is talc.

❖ Any ten numbers may be used to represent the categories, provided they are ordered /sorted in ascending/descending magnitude.

Handwritten notes:
Softest ← 10 Talc 3 4 . . . 1 most hardest Diamond
M3 M4 M5
3 4 5

❖ For instance, the integers 1–10 would be natural to use.

❖ However, any sequence of increasing numbers may also be used.

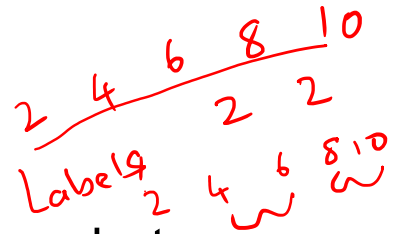
Sl. No.	College	Rank
1	X Arts & Science Institute	1
2	Y College of Music	2
3	Z College of Fashion Technology	3
4	S College of Engineering	4

- ❖ The best, second best, etc. are obviously evident
- ❖ However, the difference between the first and second, the second and third, etc. may not be established
- ❖ We see order, but not degree of difference within ordinal data.

Types of Variables: Ordinal

- ❖ Often investigators classify people, or ask them to classify themselves, along some continuum
- ❖ For example, a physician may classify a patient's disease status as none = 1, mild = 2, moderate = 3, and severe = 4.
- ❖ Clearly, increasing numbers indicate increasing severity, but
- ❖ It is not clear that the difference between not having an illness and having a mild case is the same as between having a mild case and a moderate one.
- ❖ Hence, this is an ordinal variable.

Types of Variables: Interval



❖ An interval variable is a variable in which the differences between successive values are always the same.

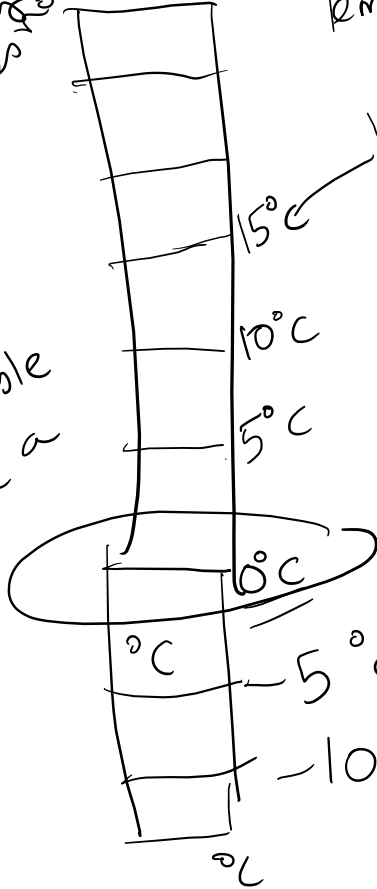
❖ For e.g., the variable “temperature,” in degrees Celsius, is measured on the interval scale since the difference between 12°C and 13°C is the same as the difference between 13°C and 14°C or the difference between any two successive temperatures.

Temperature
[°C] K, °F

Currency
 0 No Money
 Money does not exist

Interval variable
 You do not have a
 universal zero

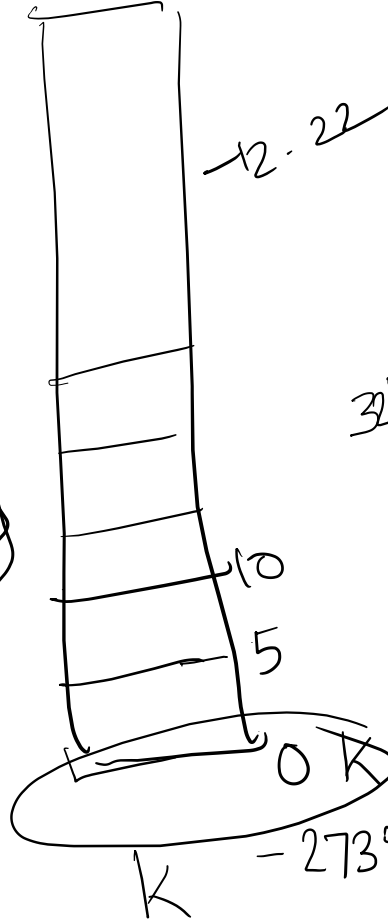
Temp. scale



12.22
 $\frac{10^\circ\text{F}}{5^\circ\text{F}} = 2$

$\frac{10^\circ\text{C}}{5^\circ\text{C}} = 2$

$\frac{-12.2^\circ\text{C}}{-15^\circ\text{C}} = 0.81$



12.22

32°F

0°F

-5°F

-273°C

$$\frac{F - 32}{9} = \frac{C}{5}$$

$$^\circ\text{C} = 0$$

$$^\circ\text{F} = 32$$

$$^\circ\text{K} = 273$$

$$-32 \times 5 = ^\circ\text{C}$$

$$\frac{-160}{9} = -17.8$$

$$-18^\circ\text{C} \equiv 0^\circ\text{F}$$

Types of Variables: Interval

- ❖ In contrast, the Mohs Hardness Scale does not satisfy this condition since the intervals between successive categories are not necessarily the same.
- ❖ The scale must satisfy the basic empirical operation of preserving the equality of intervals
- ❖ But we can compute average values, standard deviations, or correlations for interval-scaled data.

Types of Variables: Interval

Sl. No.	Month	Temperature (°C)
1	February	30
2	May	45
3	September	36
4	December	25

- ❖ Interval scaled data are more informative than nominal or ordinal scaled data, since numbers relate to the quantity.
- ❖ Two interval variable type quantities may be directly compared to assess the extent of their difference.
- ❖ Example: Temperatures are interval variables.
- ❖ Temperature in May is 15 °C hotter than in February.
- ❖ In December, the temperature is 11 °C colder than in September.

*Any further e.g. of
Interval Variables?*

Types of Variables: Ratio

- ❖ Ratio variables are interval variables with a reference to “nothing” given by zero .

$$\frac{5^{\circ}\text{C}}{10^{\circ}\text{C}} \neq \frac{5^{\circ}\text{F}}{10^{\circ}\text{F}} \quad \frac{1}{2} = \frac{5\text{ cm}}{10\text{ cm}} = \frac{5\text{ m}}{10\text{ m}}$$

- ❖ For example, height is a ratio variable as 0 height in any dimension is a naturally defined and identified point on the scale.
- ❖ We may change the unit of measurement (e.g., cm to feet), but we would still save the zero point and the ratio of any two values of height irrespective of dimensions.

Types of Variables: Ratio

- ❖ Temperature is not a ratio variable as we may choose the zero point arbitrarily, thus not preserving ratios.
- ❖ 0°C is not same temperature as 0°F.
- ❖ Ratio-scaled data permits all arithmetic operations as well as the calculation of all statistical measures

Types of Variables: Ratio

- ❖ Ratio scaled data convey the greatest amount of information, as '0' has meaning.
- ❖ Value of zero indicates complete absence of the entity being measured.
- ❖ With a true zero in your scale, you can calculate ratios of values.
- ❖ E.g., you can say that 4 children is twice as many as 2 children in a household.
- ❖ Similarly, 8 years is double 4 years of experience.
- ❖ Business variables measured in \$, such as revenues and profits, as well as those measured in percentages, such as market share, are ratio scaled variables.
- ❖ A value of zero means no revenue, no profits, or no market share.

Types of Variables: Binary & Dummy

- ❖ Variables that comprise only **2** categories are termed dichotomous
- ❖ Special case of nominal variables. For e.g.:  YES/NO decisions (e.g., purchase/non-purchase), positive/negative test results (e.g., Corona-positive/Corona-negative), question of survival (e.g., patient survived/patient died), tossing a coin (head/tail), pass/fail
- ❖ Dichotomous variables are mutually exclusive, occurrence of one precludes the occurrence of the other

Types of Variables: Binary & Dummy

- ❖ Occurrence of one instance of the binary dichotomous variable precludes the occurrence of the other – If YES, ~~NO is ruled out~~.
- ❖ If dichotomous variables are coded as 0 and 1, they are called **binary** variables or **dummy** variables.
- ❖ Binary variables may be treated like metric variables.

Types of Variables: Binary & Dummy

- ❖ The mean value of a binary variable indicates the proportion the attribute value coded with 1 occurs in a data set

$$\frac{\sum_{i=0}^n \text{number of ones}}{n}$$

Sample

0 1
2 options
2-1 option

Mean value = $\frac{3}{5}$
= 0.6

- ❖ E.g., 'purchase' is coded with 1 and 'no-purchase' with 0, the mean value of 0.75 means that the product was purchased by 75% i.e., by 30 out of the 40 respondents.

Types of Variables: Dummy

- ❖ Nominal variables with more than two categories may not be treated like metric variables.
- ❖ But it is possible to replace a nominal variable say “green” among three colors “red”, “green” and “blue” with several dummy variables that may then be interpreted like metric variables (say

1 0)

Handwritten notes:

$\left\{ \begin{array}{l} \text{Green} \\ \text{Blue} \\ \text{red} \end{array} \right.$ $\begin{array}{c} 1 \\ 0 \\ 0 \end{array}$ $\begin{array}{c} 0 \\ 1 \\ 0 \end{array}$ $\leftarrow$ Reference

k categorical levels
 $k-1$ dummy / indicator variables

Types of Variables: Dummy

- ❖ Supplier wants to know if customer choice is linked to color of packing
- ❖ Three packing colors red, yellow and green are considered
- ❖ The nominal variable 'color' may be represented by three dummy variables, each of which may have the value 1 (color present) or 0 (otherwise).

Constituents of each dummy variable

two

1	0	— DV1
0	1	— DV2
0	0	— DV3

two

- ❖ For color red, for example, the coding of the dummy variable q_1 is as follows:

$$q_1 = \begin{cases} 1 & \text{if color = red} \\ 0 & \text{otherwise} \end{cases}$$

** ordinal months ?*
** * term for 01 elements within each DV.?*

Types of Variables: Dummy

- ❖ Similarly, a dummy variable q_2 maybe defined for yellow and a dummy variable q_3 for green.

0 0
0 1
1 0

0 0 1
0 0 0

- ❖ If only these three colors are considered, one of the three dummies becomes redundant and unnecessary: if $q_1 = 0$ and $q_2 = 0$, q_3 must be equal to 1.

- ❖ The three colors can therefore be described by two dummy variables q_1 and q_2

D1

DV1 0 1 Yellow
DV2 1 0 Green

- ❖ The third—unused—variable serves as the reference category.

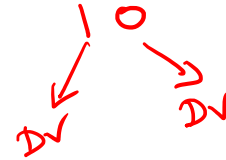
DV3 0 0 Red Ref.

Types of Variables: Dummy

- ❖ The choice of the reference category is arbitrary.
- ❖ In general, a nominal variable with n categories may be replaced with $(n-1)$ dummy variables
- ❖ Dummy variables representation is different from one – hot encoding representation

Handwritten notes for the first bullet point:
dV
Red Yellow green
1 0 0
0 1 0
0 0 1
one hot encoding
1 0 0
0 1 0
0 0 1

Handwritten notes for the second bullet point:
3 colors
2 dummy variables



Handwritten note: Dummy

Handwritten note: one hot encoding

Handwritten note: Correct the Notes.

Handwritten note: Clarify these in next class

Dummy variables and one-hot encoded variables are both techniques used to convert categorical text data into numbers, but they have a key mathematical difference: **One-hot encoding creates a new column for every category, while dummy encoding creates $N - 1$ columns** (where N is the total number of categories). [Data Science Stack Exchange +2](#)

⇒

	d1	d2	d3
1	1	0	0
	0	1	0
	0	0	1

A simple breakdown illustrates how they handle a color category with 3 options: Red, Blue, and Green.

The Core Differences

Feature	One-Hot Encoding	Dummy Encoding
Number of columns created	All N categories are represented by N binary columns. <i>0 1 0 Binary Columns</i>	$N - 1$ columns are created. One category is omitted and serves as the baseline. <i>binary</i> <i>[0 0] one color</i>
Example for Colors (Red, Blue, Green)	Requires 3 columns: <i>Color_Red</i> , <i>Color_Blue</i> , <i>Color_Green</i> . <i>1 0 0</i> <i>0 1 0</i> <i>0 0 1</i>	Requires 2 columns: <i>Color_Blue</i> , <i>Color_Green</i> . (Red is represented by <code>[0, 0]</code>). <i>1 0</i> <i>0 1</i> <i>0 0 Reference</i>

Difference Between One-Hot Encoding and Dummy Variables

Types of Variables: Indicator

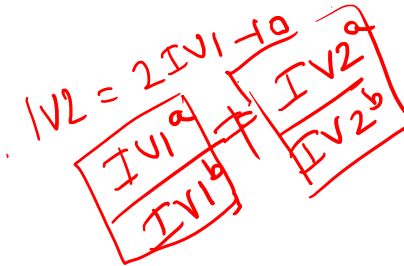
- ❖ The usual method of representing the different levels of a qualitative variable is by using indicator variables.
- ❖ For example, to introduce the effect of two different machines into a regression model, we could define an indicator variable as follows:

$x = 0$ if the observation is from machine 1

$x = 1$ if the observation is from machine 2

Partial Least Squares
Regression

- ❖ In general, a qualitative variable with t levels is represented by $t-1$ indicator variables, which are assigned the values either 0 or 1.



Traffic	8	9	Signal
Vehicles	150	250	8 2 min
IV1		IV2	4 min
# of Vehicles		# Waiting time	

Types of Variables: Indicator & Dummy

- ❖ Thus, if there were three machines, the different levels would be accounted for by two indicator variables defined as follows:

x_1	x_2	
0	0	if the observation is from machine 1
1	0	if the observation is from machine 2
0	1	if the observation is from machine 3

Indicator variables are also referred to as dummy variables.

Minor Semantic Distinction

While the terms are synonymous, statisticians sometimes make a slight distinction in how they refer to **groups**:

- **Indicator Variable:** Typically used to describe an isolated binary condition (e.g., Yes/No).
- **Dummy Variable:** Often used when a single categorical variable has 3 or more levels and must be broken down into a set of indicators. For example, a "Day of the Week" variable would be represented by a set of dummy variables (e.g., Monday = 1, others = 0).

Types of Variables: Discrete and Continuous

- ❖ Many authors use the term categorical to refer to nominal and ordinal variables
- ❖ Variables may be classified as discrete or continuous.
- ❖ A variable is called continuous if it can take on any value in a specified range.
- ❖ Thus, the height of an individual may be 70 or 70.4539 inches.
- ❖ Any numerical value within a certain range is a possible height.

Types of Variables: Discrete and Continuous

- ❖ A variable that is not continuous is discrete.
- ❖ A discrete variable takes on only certain specified values but nothing between the chosen ones are considered
- ❖ For e.g., counts are discrete variables since only 0 or positive integers are allowed.
- ❖ **In fact, all nominal and ordinal variables are discrete.**
- ❖ Interval and ratio variables may be continuous or discrete.

Types of Variables: Discrete and Continuous

- ❖ Nominal & ordinal are non-metric/categorical scales
- ❖ Interval and the ratio are metric or cardinal scales
- ❖ Scale decides information content of the data & permissible math and statistical operations
- ❖ Nominal scale is the lowest & ratio scale the highest in terms of information content
- ❖ Permissible arithmetic operations or statistics that are permissible on a lower scale are also permissible on a higher scale, but not vice versa

Types of Variables: Discrete and Continuous

- ❖ Superior the measurement level, the more is the information content of the data
- ❖ More arithmetic and statistical operations may be performed on the data.
- ❖ Possible to transform data from a higher to lower scale, but not vice versa.
- ❖ May be useful for increasing the clarity of the data or for simplifying the analysis
- ❖ For example, in many cases income classes or price classes are formed.
- ❖ This may involve transforming the originally ratio-scaled data to an interval, ordinal or nominal scale.
- ❖ Of course, transformation to a lower scale always involves a loss of information

Types of Variables: Discrete and Continuous

Type of Variable	Constituent	Permitted Math/Stat Operations
Categorical	Nominal	Frequency, Mode
	Ordinal	Median, Quantiles
Metric	Interval	Difference, Average, Standard Deviation, Correlation
	Ratio	Total, ratio, multiplication, geometric and harmonic averages, and coefficient of variation

1. Backhaus K, B Erichon, S. Gensler, R. Weiber, T. Weiber, Multivariate Analysis – An Application Oriented Introduction, 2nd ed., Springer Gabler, 2023